Functional & Non-functional Requirements

Functional and non-functional requirements

- **Functional requirements**
  - Statements of services the system should provide, how the system should react to particular inputs and how the system should behave in particular situations.
- **Non-functional requirements**
  - constraints on the services or functions offered by the system such as timing constraints, constraints on the development process, standards, etc.
- **Domain requirements**
  - Requirements that come from the application domain of the system and that reflect characteristics of that domain

Functional Requirements

Describe functionality or system services

- **Depend on the type of software**, expected users and the type of system where the software is used

- **Functional user requirements may be high-level statements of what the system should do BUT** functional system requirements should describe the system services in detail

Examples of functional requirements

- The user shall be able to search either all of the initial set of databases or select a subset from it.
- The system shall provide appropriate viewers for the user to read documents in the document store.
- Every order shall be allocated a unique identifier (ORDER_ID) which the user shall be able to copy to the account's permanent storage area.

Non-functional requirements

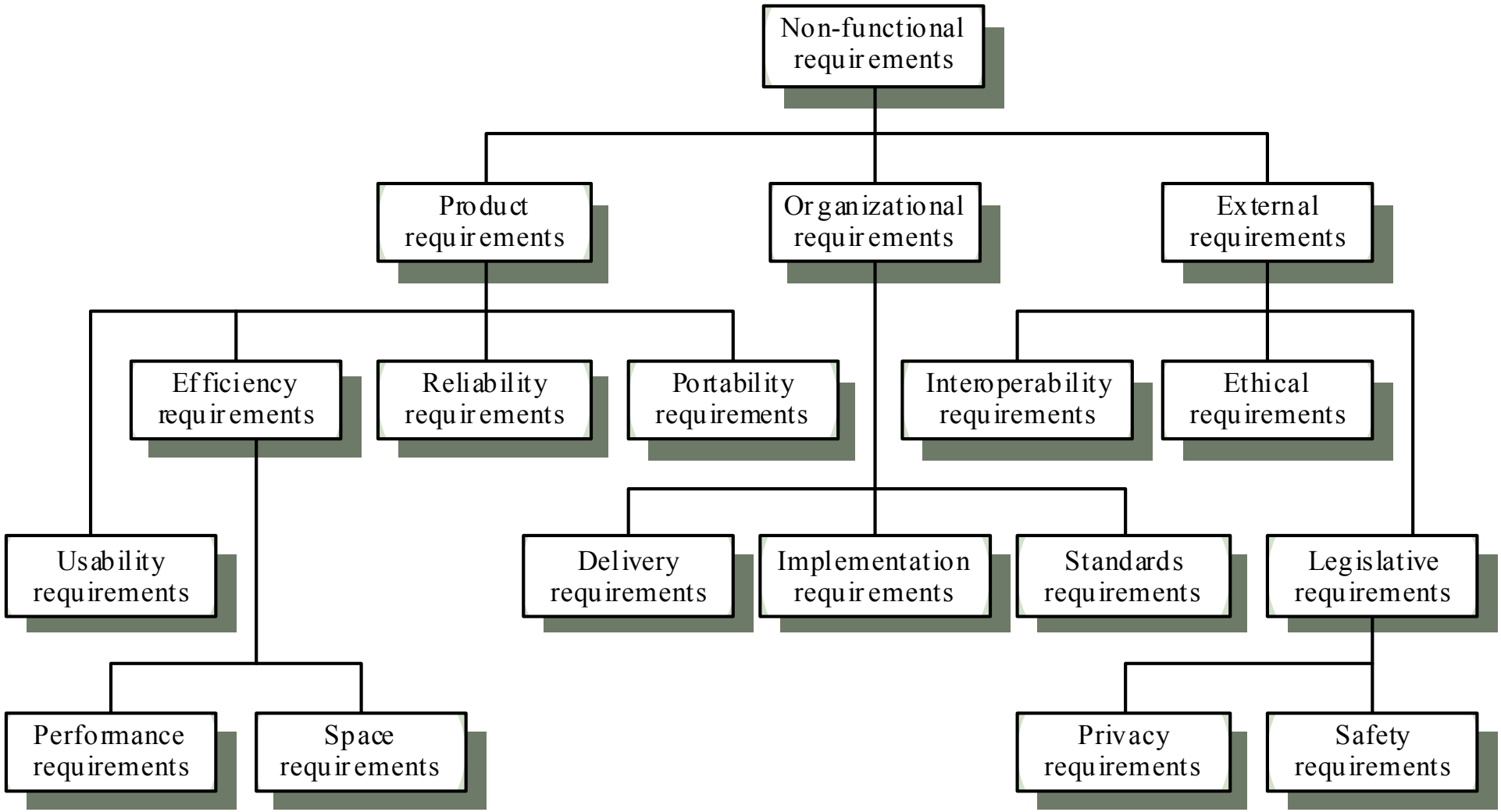
Define system properties and constraints e.g. reliability, response time and storage requirements. Constraints are I/O device capability, system representations, etc.

- **Process requirements** may also be specified mandating a particular CASE system, programming language or development method
- **Non-functional requirements** may be more critical than functional requirements. If these are not met, the system is useless

Non-functional classifications

- **Product requirements**
  - Requirements which specify that the **delivered product must behave in a particular way** e.g. execution speed, reliability, etc.
- **Organisational requirements**
  - Requirements which are a **consequence of organisational policies and procedures** e.g. process standards used, implementation requirements, etc.
- **External requirements**
  - Requirements which arise from factors which are **external to the system and its development process** e.g. interoperability requirements, legislative requirements, etc.

Non-functional requirement types



Requirements measures

Property	**Measure**
Speed	Processed transactions/second User/Event response time Screen refresh time
Size	K Bytes Number of RAM chips
Ease of use	Training time Number of help frames
Reliability	Mean time to failure Probability of unavailability Rate of failure occurrence Availability
Robustness	Time to restart after failure Percentage of events causing failure Probability of data corruption on failure
Portability	Percentage of target dependent statements Number of target systems